

SEFI–DEFI–GWFM Unified Geometric QEC Framework and the SEFI-PY Geometric Quantum Error Correction Engine

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This document provides a concise overview of the SEFI–DEFI–GWFM unified geometric field and quantum error correction framework, along with the SEFI-PY geometric QEC engine. These constructions model stability, measurement behavior, and worldline geometry using deterministic geometric principles directly relevant to quantum readout engineering, geometric QEC, and systems-level quantum architecture.

I. SUMMARY

I develop a unified geometric field and quantum error correction framework (SEFI, DEFI, GWFM) and a geometric QEC engine (SEFI-PY) that models stability, measurement behavior, and worldline geometry. The work provides deterministic interpretations of quantum readout, geometric error correction, and quantum-classical interface architecture.

II. CORE CONTRIBUTIONS

- **Unified SEFI–DEFI–GWFM Framework:** Geometric field/QEC construction integrating stability, curvature, torsion, and worldline behavior.
- **SEFI-PY Geometric QEC Engine:** Deterministic stabilizer logic, geometric syndrome modeling, and curvature/torsion analysis.
- **Geometric Readout Interpretation:** Stability-driven measurement modeling aligned with quantum readout engineering.
- **Systems-Level Quantum Architecture:** Supports quantum-classical interface design, stability surfaces, and geometric QEC pathways.

III. MANUSCRIPTS AND SUBMISSION STATUS

- **AIP Advances (submitted):** “Photonic Quantum Error Correction from Geometric Stability Principles in SEFI/GWFM” Tracking #ADV26-AR-04526-TR

- **Journal of Mathematical Physics (under review):** “Geometric Worldline Foundations of the Electron Field” Tracking #JMP26-AR-02470
- **Journal of Mathematical Physics (under review):** “SEFI: The Single Entity Field Interpretation — Formal Mathematical Construction” Tracking #JMP26-AR-02501

IV. SEFI QEC STACK REPOSITORY

A companion geometric QEC software stack implementing stabilizer logic, geometric syndrome modeling, and worldline curvature/torsion analysis is available at: github.com/JasonSEFIDEFI/SEFI.QEC.Stack

V. LINKS

LinkedIn: linkedin.com/in/your-profile
GitHub (SEFI-PY): github.com/JasonSEFIDEFI/SEFI
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VI. ALIGNMENT WITH MICROSOFT QUANTUM

- Quantum readout stability
- Geometric quantum error correction
- Measurement interpretation
- Quantum-classical interface architecture
- Systems-level quantum design